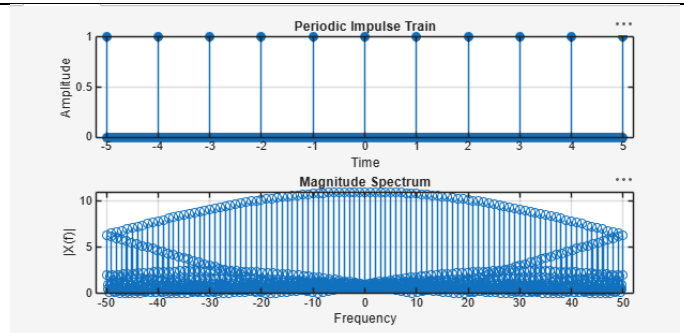


## 8.FOURIER TRANSFORM AND FREQUENCY DOMAIN ANALYSIS

### Experiment 8.1

#### Fourier Transform of Periodic Impulse Train

```
% Fourier Transform of Periodic  
Impulse Train  
clc;  
clear;  
close all;  
% Time axis  
t = -5:0.01:5;  
% Generate periodic impulse train  
x = zeros(size(t));  
x(1:100:end) = 1; % Impulse every  
100 samples  
% Fourier Transform  
X = fftshift(fft(x));  
% Frequency axis  
N = length(X);  
f = linspace(-50,50,N);  
% Plotting  
figure;  
subplot(2,1,1);  
stem(t,x,'filled');  
grid on;  
title('Periodic Impulse Train');  
xlabel('Time');  
ylabel('Amplitude');  
subplot(2,1,2);  
stem(f,abs(X));  
grid on;  
title('Magnitude Spectrum');  
xlabel('Frequency');  
ylabel('|X(f)|');
```



## Experiment 8.2

### Discrete Time Fourier Transform

```
% Discrete-Time Fourier Transform (DTFT)
clc;
clear;
close all;
% Input sequence
x = [1 2 3 4];
% Frequency range
w = -pi:0.01:pi;
% DTFT computation
X = zeros(size(w));
for k = 1:length(w)
    for n = 1:length(x)
        X(k) = X(k) + x(n)*exp(-1j*w(k)*(n-1));
    end
end
% Plotting
figure;
subplot(2,1,1);
stem(0:length(x)-1,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
plot(w,abs(X),'LineWidth',2);
grid on;
title('Magnitude Spectrum of DTFT');
xlabel('\omega (rad/sample)');
ylabel('|X(e^{j\omega})|');
```

